ORYX 理工 INSTITUTE

STUDENT MODULE

ORX 1101

Tools of Measurement

Diploma in Applied Engineering

Level 5 · 12 credits · Semester 1

Name _____

Student no. _____ Class _____



WELCOME

Tools of Measurement

Module code	ORX 1101
Level and credits	Level 5, 12 credits (120 notional hours)
Delivery	Weekly workshop and supervised practical
Prerequisites	None

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What you will be able to do

- Choose the right instrument for a given measurement task.
- Read and record measurements to the correct precision.
- Explain and reduce common sources of error.
- Report results so that another person can repeat them.

STUDY GUIDE

How to use this module

Each unit follows the same four steps.

Step	What you do	Time
Read	Read the unit text and study the figures.	30 minutes
Do	Complete the activity at your bench.	60 minutes
Check	Answer the review questions.	20 minutes
Record	Enter your results in your logbook.	10 minutes

> **TIP**
> Keep your logbook open while you read. A result you do not record on the day is a result you do not have.

Weekly plan

Week	Unit	Focus
1 to 3	Unit 1	Why we measure
4 to 6	Unit 2	Instruments
7 to 12	Practical	Logbook and final examination

UNIT 1

Why we measure

IN THIS UNIT
You will learn why measurement matters, follow the five-step measurement cycle, and tell precision from accuracy.

1.1 Numbers in the trades

Every trade begins with a number. A carpenter cuts to a length, an electrician checks a voltage, a nurse reads a dose. When the number is wrong, the work is wrong, however good the hands that did it.

1.2 The measurement cycle

Measuring well is a habit with five steps. You define what you need to know, measure it, record it, check it, and report it. When the result cannot be repeated, you start again.

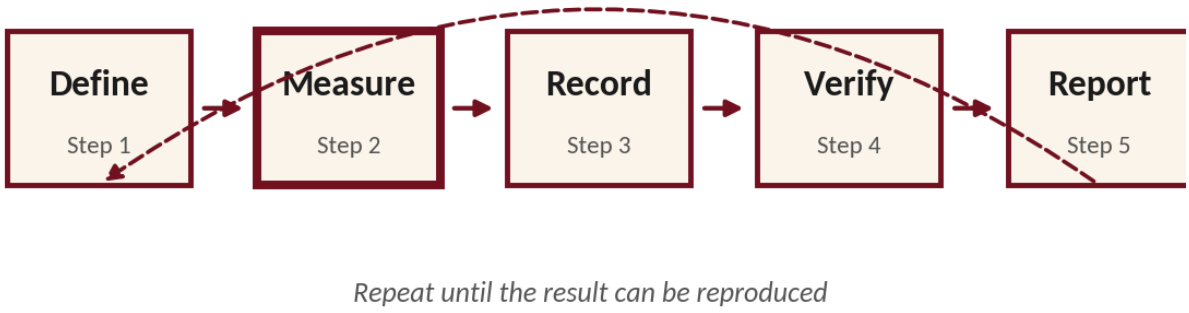


Figure 1.1. The measurement cycle.

KEY TERM
Precision describes how closely repeated measurements agree with each other. Accuracy describes how close they are to the true value.

1.3 Activity

Measure the length of your workbench three times, using a different instrument each time. Record each result and explain any differences.

Unit summary

- Measurement turns a question into a number you can repeat.
- The cycle has five steps: define, measure, record, verify, report.
- Precision is agreement; accuracy is closeness to the truth.

Review questions

1. Name the five steps of the measurement cycle.

2. Explain the difference between precision and accuracy, using an example from your own trade.

UNIT 2

Instruments

Choosing an instrument means matching its precision to the task. A steel rule is enough for a rough cut; a calliper is needed for a bearing seat.

Instrument	Measures	Typical precision
Steel rule	Length	1 mm
Vernier calliper	Length, diameter	0.02 mm
Micrometer	Thickness	0.01 mm
Multimeter	Voltage, current	Depends on range

> **REMEMBER**
> An instrument cannot give you more precision than it was built with.

Activity 2.1

Choose an instrument for each of three tasks in your workshop. Explain your choice.

ASSESSMENT

Assessment

Component	Weight	What is assessed
Practical logbook	30%	Complete, dated, repeatable records
Workshop test	20%	Instrument choice and reading
Final practical examination	50%	The full measurement cycle on a set task

Marking guide

Criterion	Excellent	Competent	Not yet
Records	Complete	Mostly complete	Missing
Precision	Correct	Minor errors	Incorrect
Reporting	Repeatable	Clear	Unclear

SUBMISSION

Hand in your logbook at the start of Week 12. Late work loses ten percent per day.

NOTES

Notes

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You are only as good as your tools. We forge them.

Oryx Polytechnic Institute
P.O. Box 1662, Windhoek, Namibia
+264 81 341 1522 · oryxinstitute.org